Data Communications

ACOE412

Lecture 5

Data Link Layer
Point-to-Point Access and Multiple Access

Spring 2009

0. Overview

In this lecture we will cover the following topics:

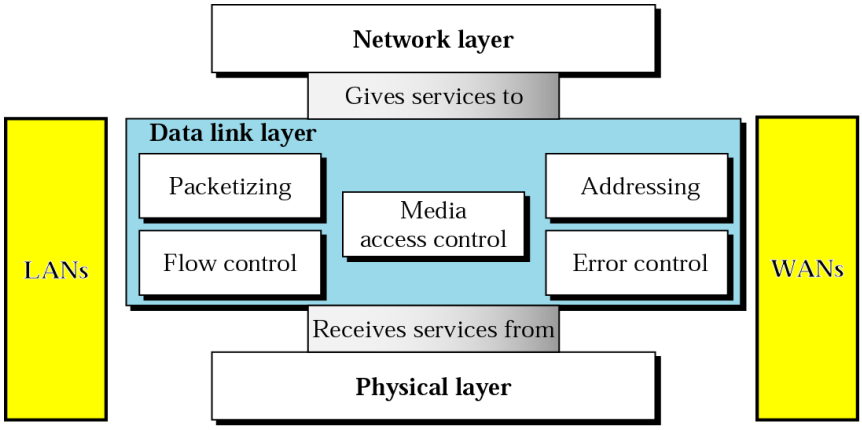
8. Data Link Layer

8.1 Types of errors
8.2 Error detection
8.3 Error correction
8.4 Summary (part 8)

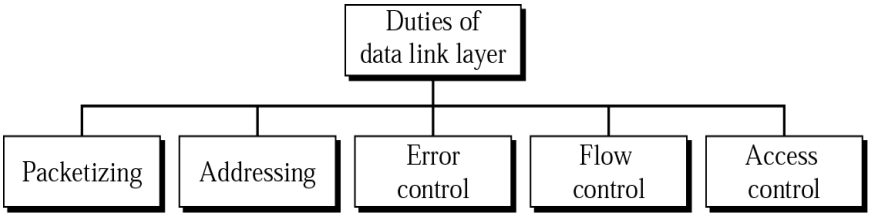
9. Point-to-Point Access & Multiple Access

9.1 Point-to-Point Access
9.2 Multiple Access
9.3 Summary (part 9)

Position of the Data-Link Layer



Data Link Layer Duties



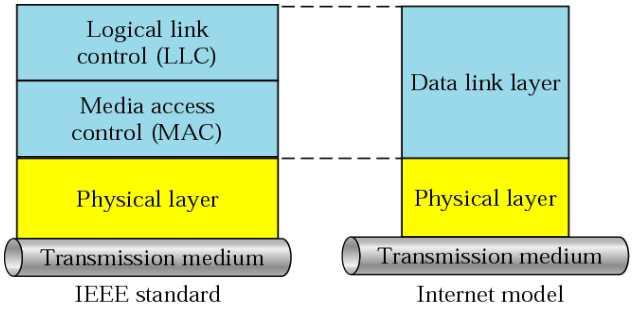
Data Link Layer Duties (1/2)

- **Packetizing:**
  - The packet coming form upper layer must be encapsulated in the appropriate packet defined by the data link layer of the under lying LAN or WAN.
  - Different protocols have different names for the packet.
  - Most LANs/WANs refer to packet as frame.
  - ATM WAN refers as Cell.
- **Addressing:**
  - The data link layer addresses are called physical addresses or MAC addresses.
  - Next-hop address is used to carry a frame across the LAN.
  - Virtual circuit address is used to carry a frame across WAN.

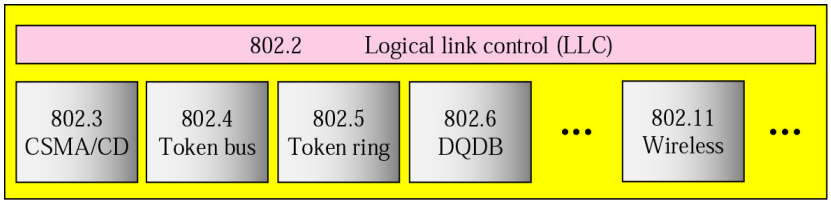
Data Link Layer Duties (2/2)

- **Error Control:**
  - Network must be able to transfer data from one device to another with complete accuracy.
- **Flow Control:**
  - The flow of data must not be allowed to overwhelm the receiver.
  - Receiving device must be able to tell the transmitting device to send few frames or stops temporarily.
- **Medium Access Control:**
  - When computers use a shared medium (cable or air), there must be a method to control the access to the medium.

LLC and MAC Sub-Layers



IEEE standards for LANs



Project 802

Note

Data can be corrupted during transmission.

Some applications require that errors be detected and corrected.

8.1 ERROR DETECTION & CORRECTION

Let us first discuss some issues related, directly or indirectly, to error detection and correction.

Topics discussed in this section:

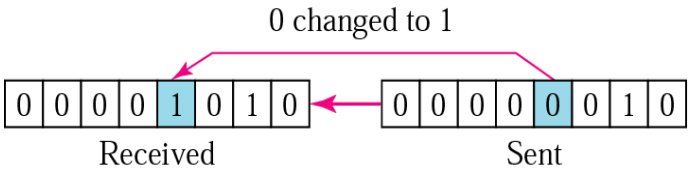
Types of Errors
Redundancy
Detection Versus Correction
Forward Error Correction Versus Retransmission
Coding
Modular Arithmetic

Note

In a single-bit error, only 1 bit in the data unit has changed.

Single-bit error

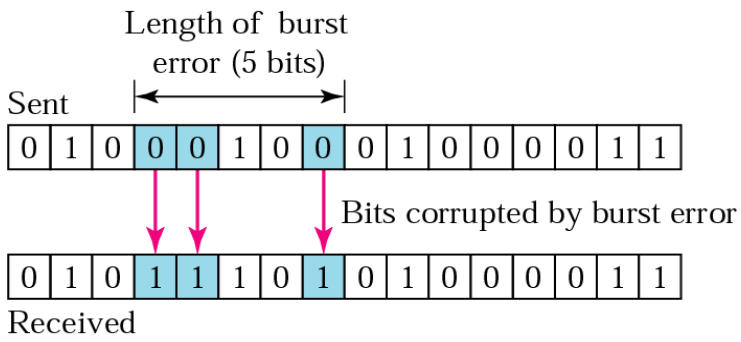
- Data can be corrupted during transmission. For reliable communication, errors must be detected and corrected.



- In a single-bit error, only one bit in the data unit has changed.

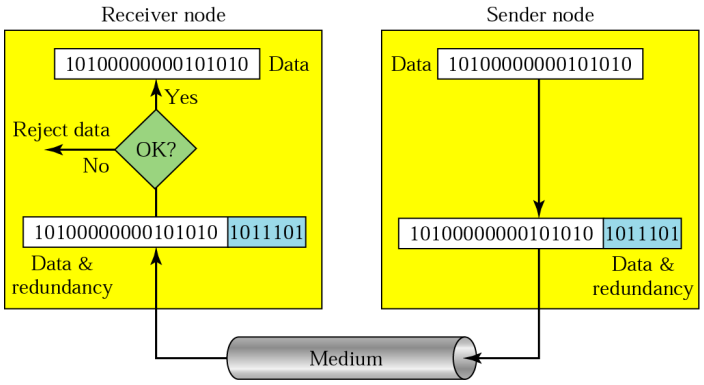
Burst error

- A burst error means that 2 or more bits in the data unit have changed.

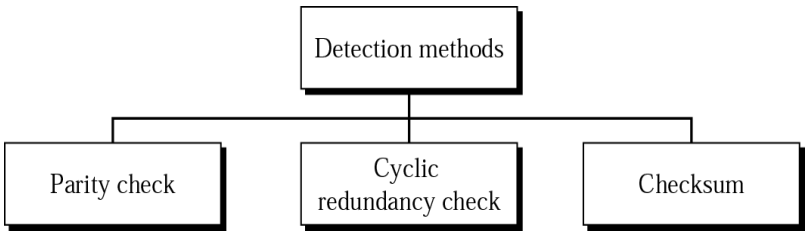


ERROR DETECTION

- Error detection uses the concept of redundancy, which means adding extra bits for detecting errors at the destination.

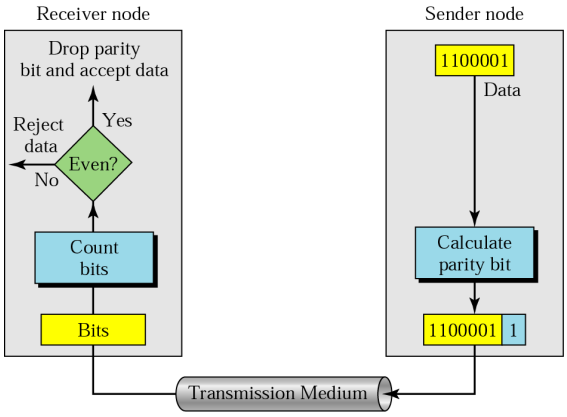


Detection Methods



Parity Check

- Simple or Two Dimensional
- In parity check, a parity bit is added to every data unit so that the total number of 1s is even (or odd for odd-parity).



Example

Suppose the sender wants to send the word *world*. In ASCII the five characters are coded as

1110111 1101111 1110010 1101100 1100100

The following shows the actual bytes sent

11101110 11011110 11100100 11011000 11001001

Example

Now suppose the word world in Example 1 is received by the receiver without being corrupted in transmission.

11101110 11011110 11100100 11011000 11001001

The receiver counts the 1s in each character and comes up with even numbers (6, 6, 4, 4, 4). The data are accepted.

Example

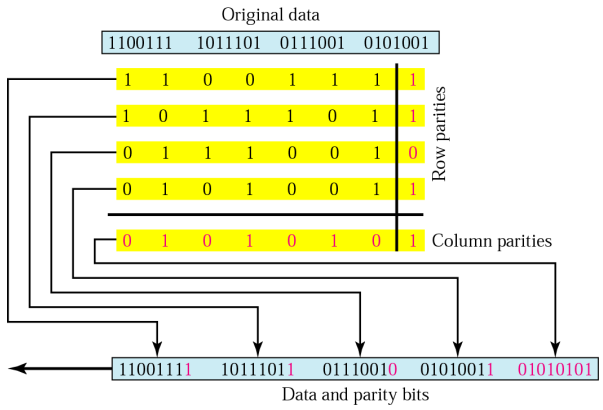
Now suppose the word world in Example 1 is corrupted during transmission.

11111110 11011110 11101100 11011000 11001001

The receiver counts the 1s in each character and comes up with even and odd numbers (7, 6, 5, 4, 4). The receiver knows that the data are corrupted, discards them, and asks for retransmission.

Two-dimensional parity

- Simple parity can detect burst errors only if the total number of errors in each data unit is odd.
- In two-dimensional parity check, a block of bits is divided into rows and a redundant row of bits is added to the whole block.



Example

Suppose the following block is sent:

10101001 00111001 11011101 11100111 10101010

However, it is hit by a burst noise of length 8, and some bits are corrupted.

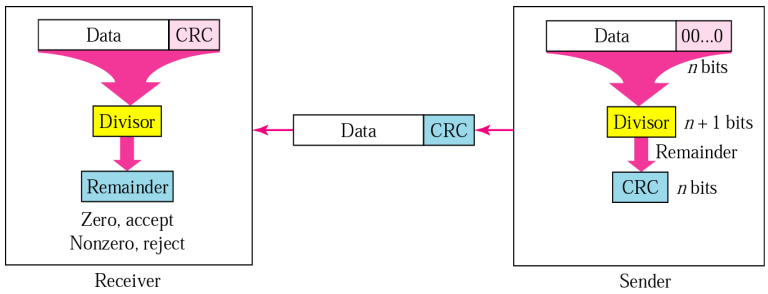
10100011 10001001 11011101 11100111 10101010

When the receiver checks the parity bits, some of the bits do not follow the even-parity rule and the whole block is discarded.

10100011 10001001 11011101 11100111 **1**0**101**0**1**0

Cyclic Redundancy Check CRC

- Uses Binary division
- Add CRC remainder to data.
- Number of 0s appended is one less than the Divisor
- Appending CRC to the end of the data must make resulting bit sequence divisible by the divisor



Binary division in a CRC generator

- Uses Modulo-2 division
- Each bit of the divisor is subtracted from the corresponding bit of the dividend without disturbing the next-higher bit.

```
                    Quotient
                    1 1 1 1 0 1
Divisor  1 1 0 1 ) 1 0 0 1 0 0 [0 0 0]  <- Data plus extra zeros
                   1 1 0 1
                     1 0 0 0
                     1 1 0 1
                       1 0 1 0
                       1 1 0 1
                         1 1 1 0
                         1 1 0 1
                           0 1 1 0
When the leftmost bit      0 0 0 0
of the remainder is zero,    1 1 0 0
we must use 0000 instead     1 1 0 1
of the original divisor.       [0 0 1]  Remainder
```


Binary division in a CRC checker

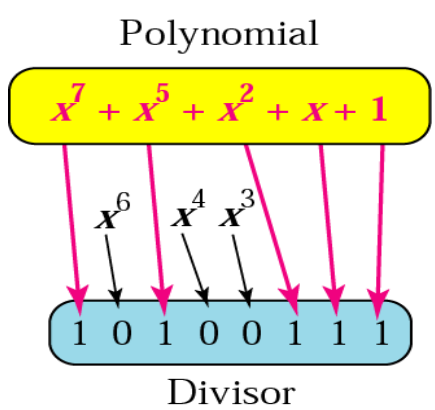
```
                    Quotient
                    1 1 1 1 0 1
Divisor  1 1 0 1 ) 1 0 0 1 0 0 0 0 1  <- Data plus CRC received
                   1 1 0 1
                     1 0 0 0
                     1 1 0 1
                       1 0 1 0
                       1 1 0 1
                         1 1 1 0
                         1 1 0 1
                           0 1 1 0
When the leftmost bit      0 0 0 0
of the remainder is zero,    1 1 0 1
we must use 0000 instead     1 1 0 1
of the original divisor.       [0 0 0]  Result
```

A polynomial

- Polynomial should not be divisible by x.
- Polynomial should be divisible by x+1.

$$x^7 + x^5 + x^2 + x + 1$$

A polynomial representing a divisor



Standard polynomials

Name	Polynomial	Application
CRC-8	$x^8 + x^2 + x + 1$	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$	ATM AAL
ITU-16	$x^{16} + x^{12} + x^5 + 1$	HDLC
ITU-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	LANs

Example

It is obvious that we cannot choose **x** (binary 10) or **$x^2 + x$** (binary 110) as the polynomial because both are divisible by x.

However, we can choose **x + 1** (binary 11) because it is not divisible by x, but is divisible by x + 1.

We can also choose **$x^2 + 1$** (binary 101) because it is divisible by x + 1 (binary division).

Example

The CRC-12

$$x^{12} + x^{11} + x^3 + x + 1$$

which has a degree of 12, will detect all burst errors affecting an odd number of bits, will detect all burst errors with a length less than or equal to 12, and will detect, 99.97 percent of the time, burst errors with a length of 12 or more.

Checksum

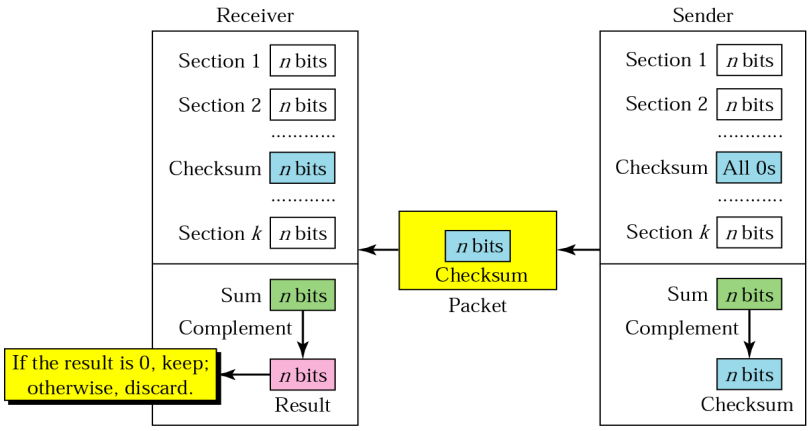
The sender follows these steps:

- The unit is divided into k sections, each of n bits.
- All sections are added using one's complement to get the sum.
- The sum is complemented and becomes the checksum.
- The checksum is sent with the data.

The receiver follows these steps:

- The unit is divided into k sections, each of n bits.
- All sections are added using one's complement to get the sum.
- The sum is complemented.
- If the result is zero, the data are accepted: otherwise, rejected.

Checksum



Example

Suppose the following block of 16 bits is to be sent using a checksum of 8 bits.

10101001 00111001

The numbers are added using one's complement

```
                10101001
                00111001
                ------------
Sum             11100010
Checksum        00011101
```

The pattern sent is 10101001 00111001 **00011101**

Example

Now suppose the receiver receives the pattern sent in Example 7 and there is no error.

10101001 00111001 00011101

When the receiver adds the three sections, it will get all 1s, which, after complementing, is all 0s and shows that there is no error.

	10101001
	00111001
	00011101
Sum	11111111
Complement	**00000000** means that the pattern is OK.

Example

Now suppose there is a burst error of length 5 that affects 4 bits.

10101**111 11**111001 00011101

When the receiver adds the three sections, it gets

	10101111
	11111001
	00011101
Partial Sum	**1** 11000101
Carry	**1**
Sum	11000110
Complement	**00111001** **the pattern is corrupted.**

Performance of Checksum

- Almost detects all errors involving odd numbers of bits or even.
- However, if one or more bits of a segment are damaged and the corresponding bit or bits of opposite value in a second segment are also damaged, then the sum of columns will not change.
- Receiver will not be able to detect the error

ERROR CORRECTION

- **Retransmission**
- **Forward Error Correction**
  - **Hamming code**
  - **Burst Error Correction**

Forward Error Correction

Data and redundancy bits

Number of data bits m	Number of redundancy bits r	Total bits m + r
1	2	3
2	3	5
3	3	6
4	3	7
5	4	9
6	4	10
7	4	11

Positions of redundancy bits in Hamming code

11	10	9	8	7	6	5	4	3	2	1
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

Calculating the Hamming Code

- The key to the Hamming Code is the use of extra parity bits to allow the identification of a single error. Create the code word as follows:
- Mark all bit positions that are powers of two as parity bits. (positions 1, 2, 4, 8, 16, 32, 64, etc.)
- All other bit positions are for the data to be encoded. (positions 3, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 17, etc.)
- Each parity bit calculates the parity for some of the bits in the code word. The position of the parity bit determines the sequence of bits that it alternately checks and skips.
- Position 1: check 1 bit, skip 1 bit, check 1 bit, skip 1 bit, etc. (1,3,5,7,9,11,13,15,...)
- Position 2: check 2 bits, skip 2 bits, check 2 bits, skip 2 bits, etc. (2,3,6,7,10,11,14,15,...)
- Position 4: check 4 bits, skip 4 bits, etc. (4,5,6,7,12,13,14,15,...)
  Position 8: check 8 bits, skip 8 bits, etc. (8-15,24-31,40-47,...)
- Set a parity bit to 1 if the total number of ones in the positions it checks is odd. Set a parity bit to 0 if the total number of ones in the positions it checks is even.

Redundancy bits calculation

r_1 will take care of these bits.

11		9		7		5		3		1
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

r_2 will take care of these bits.

11	10			7	6			3	2	
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

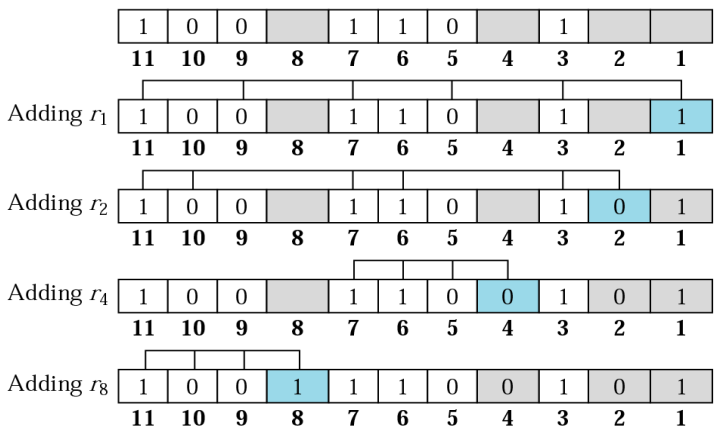
r_4 will take care of these bits.

				7	6	5	4			
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

r_8 will take care of these bits.

11	10	9	8							
d	d	d	r_8	d	d	d	r_4	d	r_2	r_1

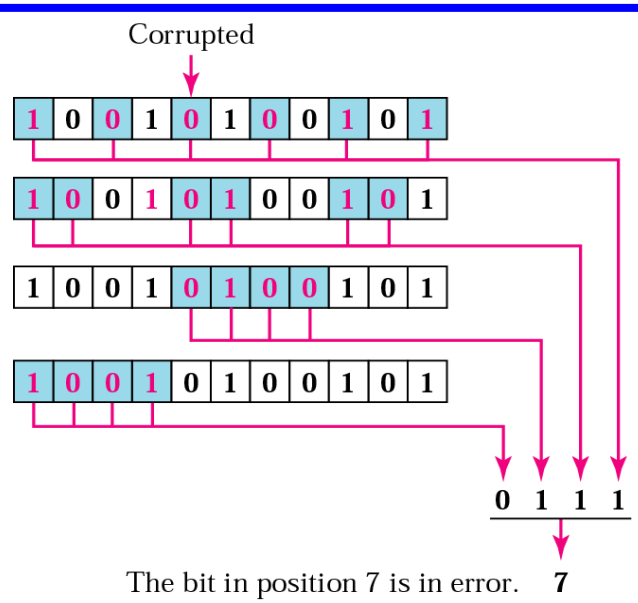
Example of redundancy bit calculation



Data:
1 0 0 1 1 0 1

Code:
1 0 0 1 1 1 0 0 1 0 1

Error detection using Hamming code

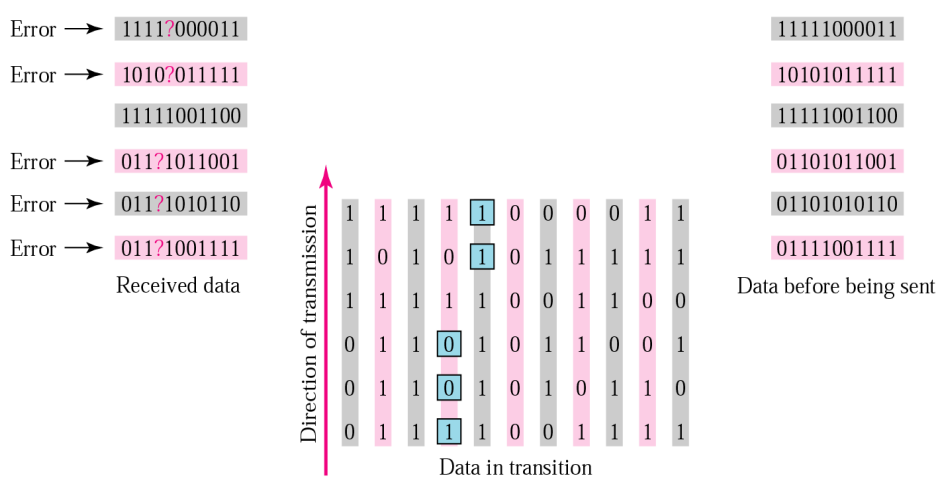


The bit in position 7 is in error.

Burst error correction

- Although the Hamming code cannot correct a burst error directly, it is possible to rearrange the data and then apply the code.
- Instead of sending all the bits in a data unit together, we can organize N units in a column and then send the first bit of each, followed by the second bit of each, and so on.
- In this way, if a burst error of M bits occurs (M<N), then the error does not corrupt M bit of one single unit; it corrupts only 1 bit of a unit.
- Organize the six data units into columns and rows. We send the first column, then the second column, and so on. The bits that are corrupted by a burst error are shown in squares. Five consecutive bits are corrupted during the actual transmission.
- However, when these bits arrive at the destination and are reorganized into data units, each corrupted bit belongs to one unit and is automatically corrected. The trick here is to let the burst error corrupt only 1 bit of each unit.

Burst error correction example



8.4 SUMMARY (part 8)

- Errors can be categorized as a single-bit error or a burst error. A single-bit error has one bit error per data unit. A burst error has two or more bit errors per data unit.
- Redundancy is the concept of sending extra bits for use in error detection.
- Three common redundancy methods are parity check, cyclic redundancy check (CRC), and checksum.
- An extra bit (parity bit) is added to the data unit in the parity check.
- The parity check can detect only an odd number of errors; it cannot detect an even number of errors.
- In the two-dimensional parity check, a redundant data unit follows n data units.
- CRC, a powerful redundancy checking technique, appends a sequence of redundant bits derived from binary division to the data unit.
- The divisor in the CRC generator is often represented as an algebraic poly-nomial.
- Errors are corrected through retransmission and by forward error correction.
- The Hamming code is an error correction method using redundant bits. The number of bits is a function of the length of the data bits.
- In the Hamming code, for a data unit of m bits, use the formula 2 r >= m +r +1 to determine r, the number of redundant bits needed.
- By rearranging the order of bit transmission of the data units, the Hamming code can correct burst errors.

9.1 POINT-TO-POINT ACCESS

In a network two devices can be connected by a dedicated or shared link. Dedicated link access is referred to as point-to-point access. Shared link access is referred to a multiple access.

Topics discussed in this section:

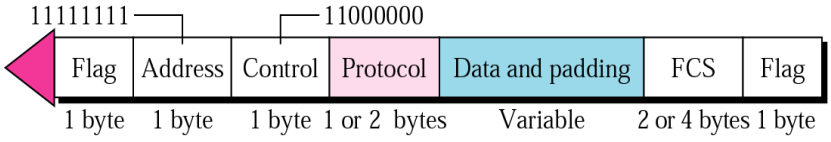
Point-to-Point Protocol (PPP)
PPP stack

PPP protocol

- Connections to ISP via traditional modem or cable modem or DSL modem requires some protocol to manage and control the transfer the data. These modems provide physical connections only.
- Today, millions of Internet users who need to connect their home PCs to the server of an Internet Service Provider (ISP) use Point-to-Point (PPP) protocol.
- **PPP Services**
  - Defines the format of the frame
  - Defines how two devices negotiate the establishment of the link and the exchange of data
  - Defines how network layer data are encapsulated in the data link frame
  - Defines how two devices can authenticate each other.

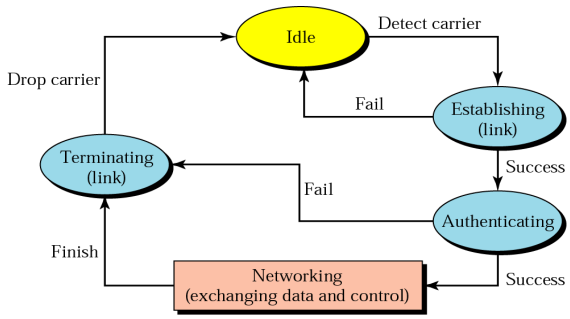
PPP Frame

- **Frame Format**
  - Flag = 01111110; Identify the boundary of PPP frame.
  - Address is always Broadcast because of point-to-point link.
  - Control field [11000000] = No sequence numbers and so no flow or error control
  - Protocol field = What is being carried in the data field: user data or other information.
  - FCS [Frame Check Sequence] = 2-byte or 4-byte CRC



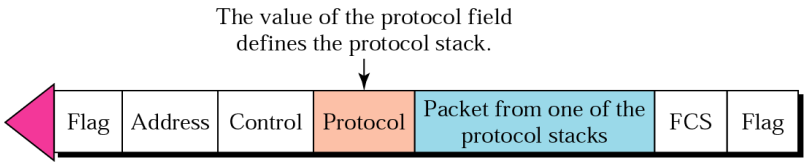
Transition States

- Idle: No active carrier, line is quiet.
- Establishing: When one endpoint starts communication, connection goes to establishing state. Negotiation of options.



Protocol Stack

- Link Control Protocol
  - Responsible for establishing, maintaining, configuring, and terminating links.
  - Negotiation mechanisms.
  - LCP occurs in Establishing or terminating state
  - LCP packets are carried in the data field of PPP frames.
  - Authentication Protocols
  - Password Authentication Protocol (PAP)
  - Challenge Handshake Authentication Protocol (CHAP)
- Network Control Protocol



Link Control Protocol (LCP)

- LCP has code, ID [identify request and reply], Length [length of entire LCP packet], Information [extra information needed for LCP packets – authentication protocol, maximum receive unit, etc.].

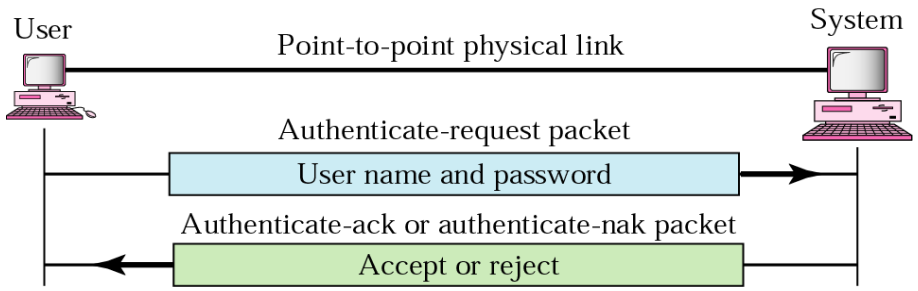
- Responsibilities of LCP
  - **Configuration packets**: Configure-request, Configure-ack, Configure-nak [negative ack certain options], Configure-reject
  - **Link Termination packets**: Terminate-request, Terminate-ack
  - **Link monitoring and debugging packets**: Code-reject, Protocol-reject, Echo-request [monitor link], Echo-reply, Discard-request [loopback by sender].

Authentication protocols

- PPP is designed for use over dial-up links where verification of user identity is necessary
- Authentication means validating the identity of a user who needs to access a set of resources
- PPP uses two protocols for authentication:
  - Password Authentication Protocol (PAP)
  - Challenge Handshake Authentication Protocol (CHAP)

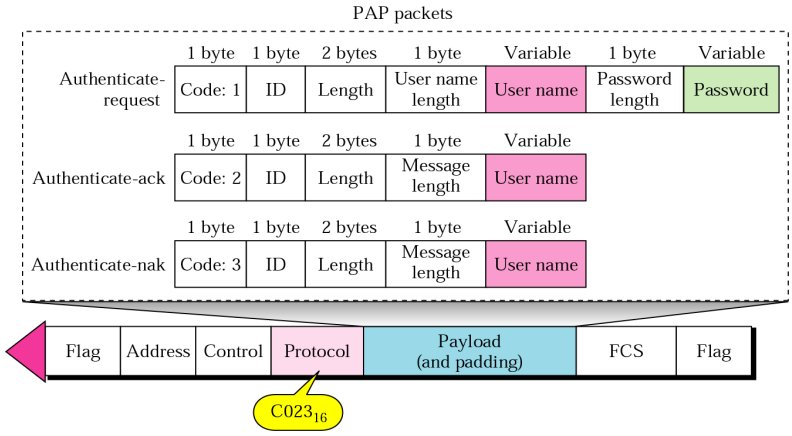
Password Authentication Protocol (PAP)

- Send user name and password
- Check validity and accept or deny.
- Access to the link can get the user name and password.



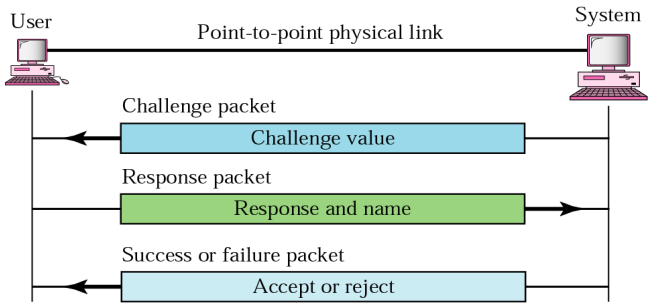
PAP packets

- PAP has protocol field value of C023 in PPP frame.
- Three PAP-packets: authenticate-request, authenticate-ack and authenticate-nak.



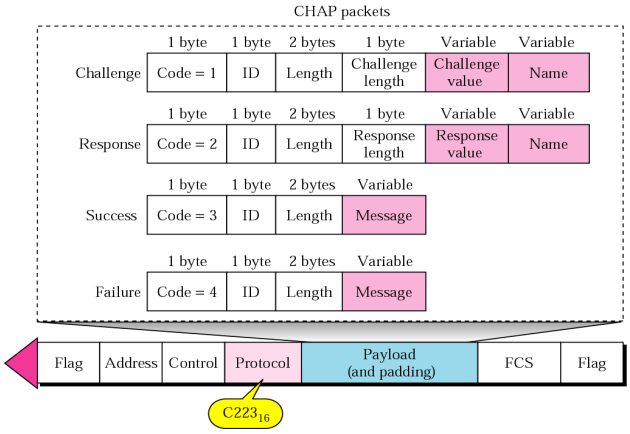
Challenge Handshake Authentication Protocol (CHAP)

- Three-way handshake. Password is never sent online.
- System sends a challenge packet containing challenge value, usually a few bytes
- User applies predefined function with challenge value, user's password. The result is sent as response.
- System does the same as user and compares the result.



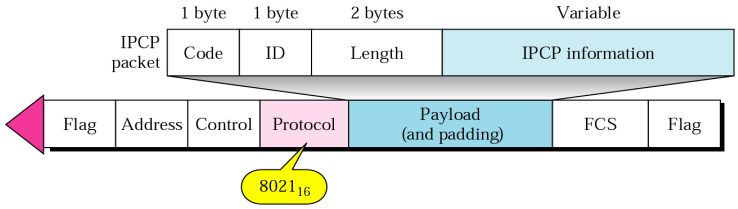
CHAP packets

- CHAP has protocol field value of C223 in PPP frame.
- Four CHAP-packets: challenge, response, success, failure.



IPCP packet encapsulated in PPP frame

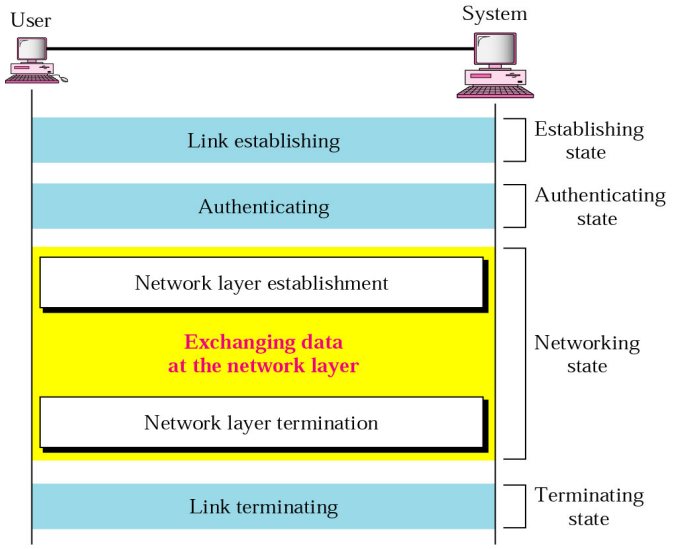
- Network Control Protocol is a set of control protocols to allow the encapsulation of data coming from network layer protocols into the PPP frame.
- The set of packets that establish and terminate a network layer connection for IP packets is called **Internetwork Protocol Control Protocol** (**IPCP**). Negotiation is done both at data link and network layers.
- Use configure-request to negotiate with the other party. Ex., to set the IP addresses and so on.
- If there is IP data, then protocol field is 0021.
- After data transfer is complete, IPCP can take control and use terminate-request and terminate-ack packets to end network connection.



Code value for IPCP packets

Code	IPCP Packet
01	Configure-request
02	Configure-ack
03	Configure-nak
04	Configure-reject
05	Terminate-request
06	Terminate-ack
07	Code-reject

Example



9.2 MULTIPLE ACCESS

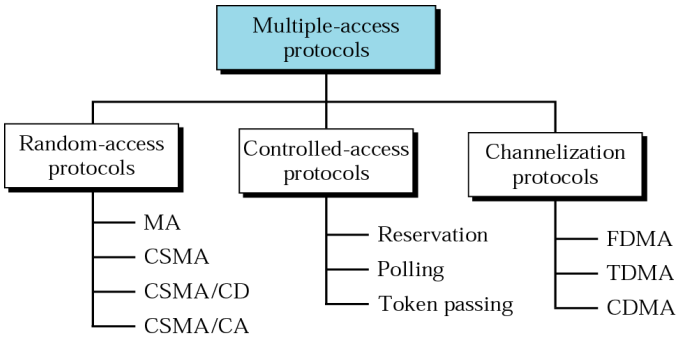
When nodes are connected to a common link, called multipoint or broadcast link, we need a multiple-access protocol top coordinate access to the link.

Topics discussed in this section:

Random Access
Controlled Access
Channelization

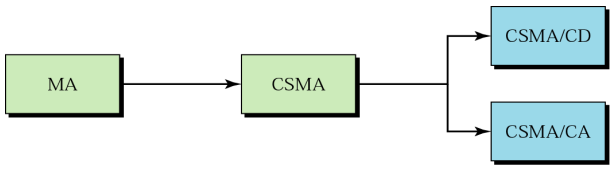
Multiple-Access Protocols

- Nodes or stations are connected to or use a common link, called a multipoint or broadcast link.
- Problem of controlling the access to the medium is similar to the rules of speaking in an assembly.



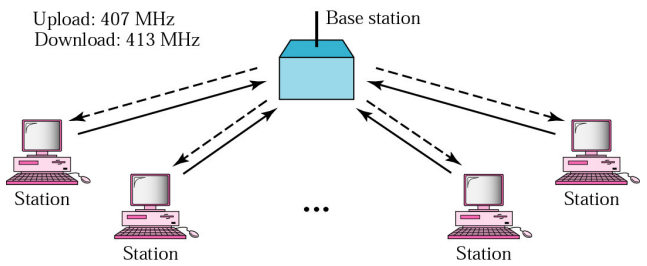
Evolution of random-access methods

- Each station has the right to the medium without being controlled by any other station.
- If more than one station tries to send, there is an access conflict (collision) and the frames will be either destroyed or modified.
  - When can the station access the medium?
  - What can the station do if the medium is busy?
  - How can the station determine the success or failure of the transmission?
  - What can the station do if there is an access conflict?

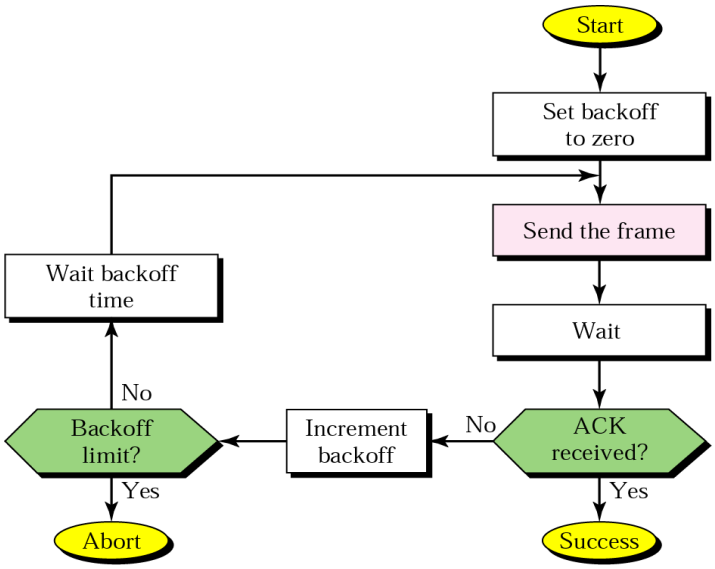


ALOHA network

- Every station that need to send a frame to another station first sends it to the base station.
- The base station receives the frame and relays it to the intended destination.
  - Multiple access
  - Acknowledgement: Expect an acknowledgement for a frame. If not received in allocated time [2*maximum propagation delay], try to send the frame again after a random amount of time.

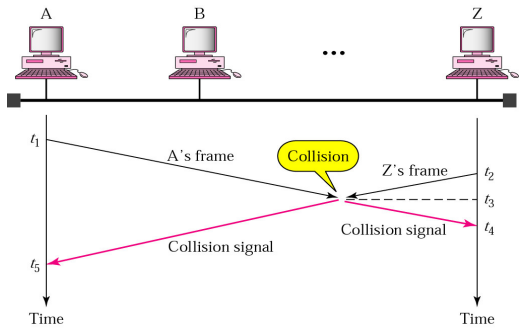


Procedure for ALOHA protocol



Collision in CSMA

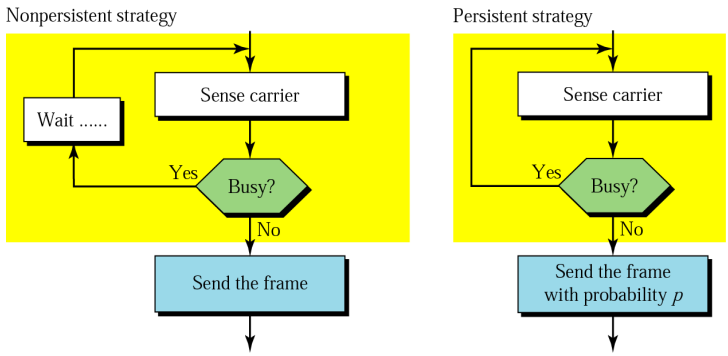
- Carrier sense multiple access
- Sense the carrier before transmit
- Possibility of collision still exists because of the propagation delay: when a station sends a frame, it takes a while (although very short) for the first bit to reach every station and for every station to sense it.



Persistence strategies

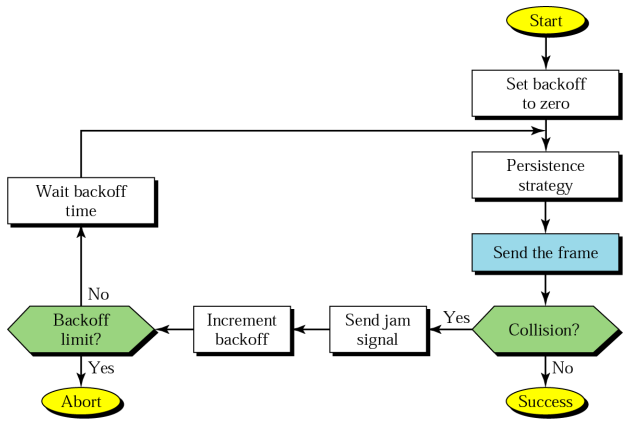
- **Nonpersistent:**
  - If a station has a frame to send, it senses the line.
  - If line is idle, the station sends the frame immediately.
  - If line is not idle, the station waits a random period of time and then senses the line again.
  - Chances of collision is reduced
  - Reduces efficiency of the network
- **Persistent:**
  - 1-Persistent: If line is idle, sends the frame immediately (with probability of 1).
  - Chances of collision is high.
  - P-Persistent: If line is idle, it may or may not send. It sends with probability p and refrains from sending with probability 1-p.

Persistence strategies



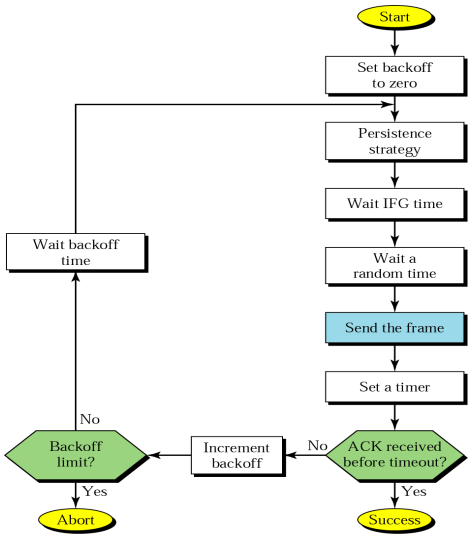
CSMA/CD procedure

- If there is collision, the frame needs to be sent again.
- To reduce the probability of collision the second time, the station waits – it needs to back off.
- Waits a little the first time, more if a collision occurs again, much more if it happens a third time, and so on.



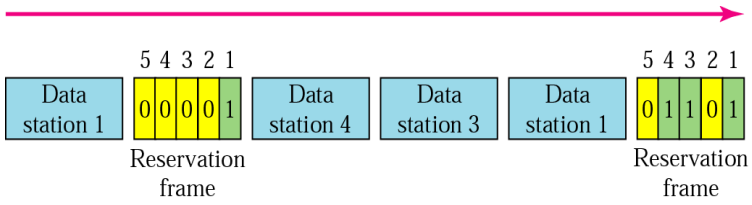
CSMA/CA Procedure

- After it finds the line idle, the station waits an IFG (Inter-frame gap) amount of time. It then waits another random amount of time. After that, it sends the frame and sets a timer.



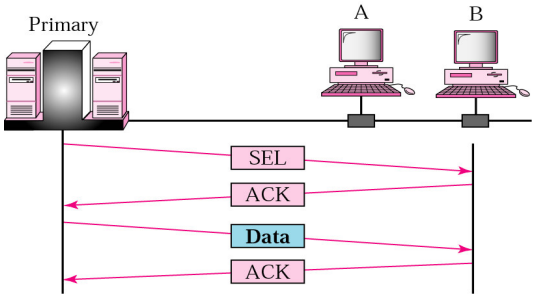
Reservation Access Method

- A station needs to make a reservation before sending data.
- Time is divided into intervals. In each interval, a reservation frame precedes the data frame sent in that interval.



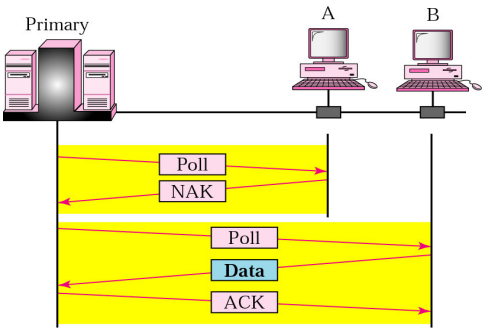
Select

- One device is designated as a primary station and the other devices are secondary stations.
- All data exchange via primary station.
- If the primary wants to receive data, it asks the secondaries if they have anything to send; this is called as polling.
- If the primary device wants to send data, it tells the secondary target to get ready to receive; this is called as selecting.



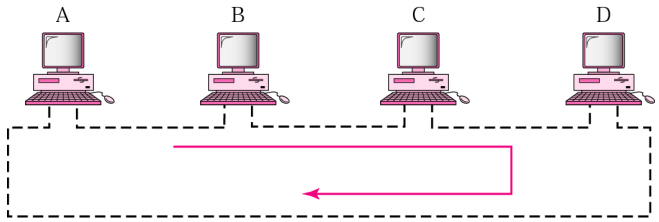
Poll

- When the primary is ready to receive data, it must ask (poll) each device in turn if it has anything to send.
- When the first secondary is approached, it responds either with a NAK frame if it has nothing to send or with data if it does.
- If the response is NAK, the primary polls the next secondary.
- If it receives data, it sends ACK frame.

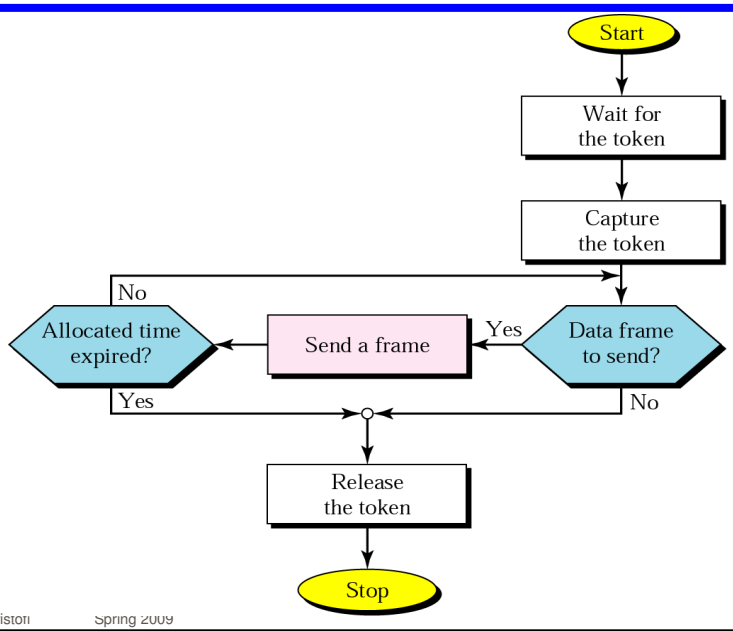


Token-passing Network

- Station is authorized to send data when it receives a special frame called a token.
- Stations are arranged around a ring.
- When no data are being sent, a token circulates the ring.
- If a station needs to send data, it waits for the token.
- The station captures the token and sends one or more frames (as long as it has frames to send or the allocated time has not expired), and finally it releases the token to be used by next station.



Token-passing procedure



Channelization

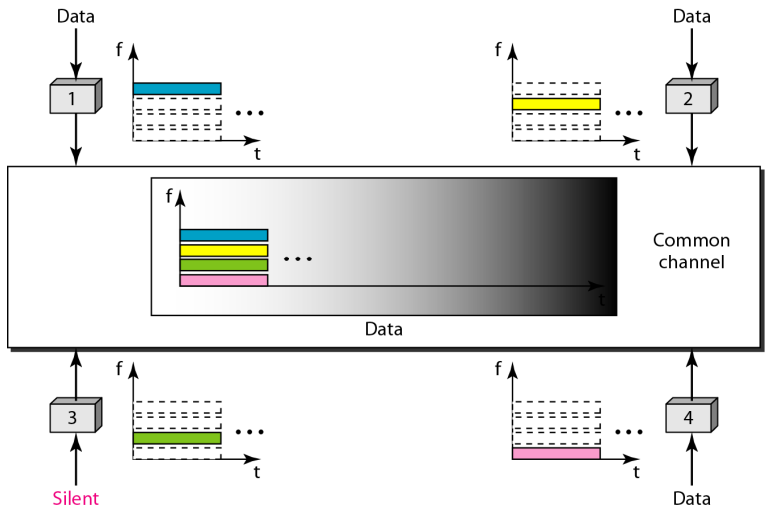
- Channelization is a multiple access method in which the available bandwidth of a link is shared in time, frequency or through code between different stations.

- In this section we will discuss the following channelization protocols:
  - **Frequency Division Multiple Access (FDMA)**
    - The total bandwidth is divided into channels.
  - **Time Division Multiple Access (TDMA)**
    - The band is divided into one channel that is time shared
  - **Code Division Multiple Access (CDMA)**
    - One channel carries all transmission simultaneously

Note

In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

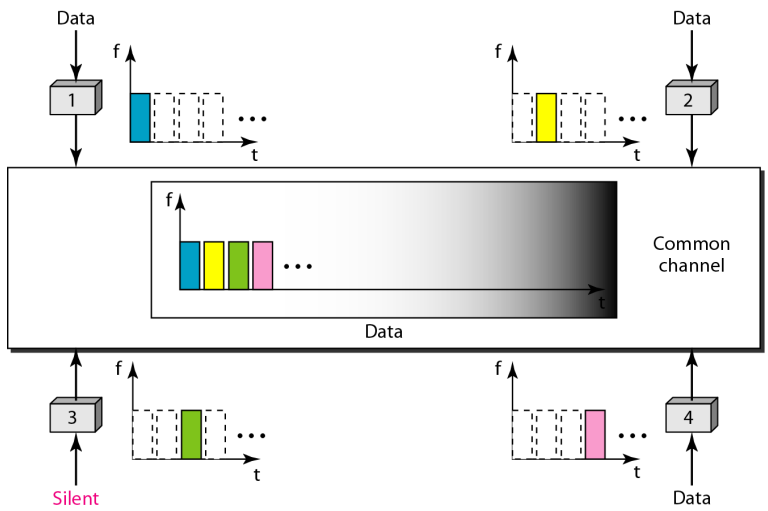
Frequency-division multiple access (FDMA)



Note

In TDMA, the bandwidth is just one channel that is timeshared between different stations.

Time-division multiple access (TDMA)



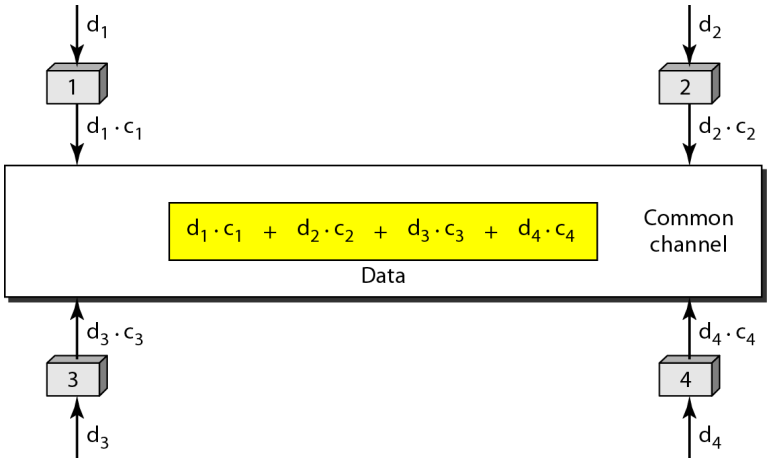
Code-division multiple access (CDMA)

CDMA is based on coding theory.
Each station is assigned a code, which is a sequence of numbers called chips.
Whenever a station needs to send data it sends an encoded data bit.

Note

In CDMA, one channel carries all transmissions simultaneously.

Simple idea of communication with code



9.3 SUMMARY (part 9)

- The Point-to-Point Protocol (PPP) was designed to provide a dedicated line for users who need Internet access via a telephone line or a cable TV connection.
- A PPP connection goes through these phases: idle, establishing, authenticating (optional), networking, and terminating.
- At the data link layer, PPP employs a version of HDLC.
- The Link Control Protocol (LCP) is responsible for establishing, maintaining, configuring, and terminating links.
- Password Authentication Protocol (PAP) and Challenge Handshake Authentication Protocol (CHAP) are two protocols used for authentication in PPP.
- PAP is a two-step process. The user sends authentication identification and a password. The system determines the validity of the information sent.
- CHAP is a three-step process. The system sends a value to the user. The user manipulates the value and sends its result. The system verifies the result.
- Network Control Protocol (NCP) is a set of protocols to allow the encapsulation of data coming from network layer protocols; each set is specific for a network layer protocol that requires the services of PPP.
- Internetwork Protocol Control Protocol (IPCP), an NCP protocol, establishes and terminates a network layer connection for IP packets.

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- W. Stallings, **Data and Computer Communications**, 7th edition, Prentice Hall, 2004
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